# Your AI Isn't Special:

## Red Teaming Before You Become a Statistic

Chase Peterson, MBA, CISSP

MinneBar 20

© 2026, Chase Peterson

# About Me

> *"I am the professor! Great and forgetful..." — Futurama*

## Academic Background

- Welding degree from St Paul College
- Mechanical Engineering degree from Michigan Tech
- MBA and MIS graduate certificate from Metro State
    - Certified Information Security Systems Professional (CISSP) and Certified in Cybersecurity (CC)
        - Cred #: 1540141

## Industry Experience

Over 15 years applying data-driven solutions across diverse sectors—from steel production to confectionary manufacturing. My roles have spanned Controls and Systems Engineering(ICS), Building Management Systems(Power and HVAC), Maintenance and Sustaining Engineering(Hydraulic).

# What We're Covering Today

1. Red Teaming vs. AI Red Teaming
2. AI Red Team Ops
3. Rapid Fire Warm-up
4. Workshop Exercise: Your Turn

# Red Teaming vs. AI Red Teaming

## Traditional Red Teaming

A structured exercise where a dedicated team attempts to breach a system's defenses — simulating a real attacker's mindset, tools, and tactics.

- Defined scope: networks, endpoints, applications
- Attack vectors are well-catalogued (MITRE ATT&CK)
- Pass/fail outcomes are often binary
- Mature tooling and established methodology

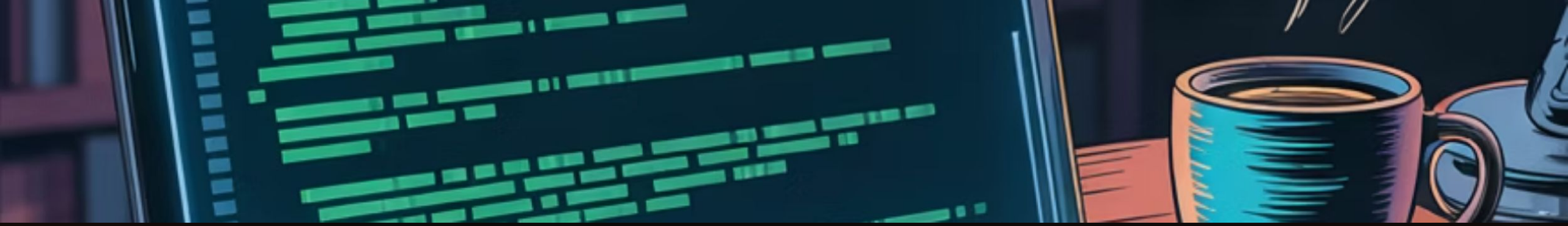
1. Define Your Threat Model
2. Map the Attack Surface
3. Structured Probing
4. Document & Triage

## AI Red Teaming

Probing of AI systems — the attack surface is probabilistic, and the harms can be invisible until they scale.

- Scope includes: prompts, training data, outputs, feedback loops
- Failures can be **subtle**: bias, hallucination, specification drift
- Harm is often distributed — no single crash event

⚠ ***The biggest risk is the quiet, compounding failure nobody flags.***



# AI Red Team Ops

**AI failures are ethical, societal  statistical, and domain-specific, the red team has to reflect that.**

## The Right Mix

- Domain experts — know what "normal" looks like
- Ethicists or policy people — catch discriminatory outputs
- Data scientists — spot distributional shift and training-data issues

https://github.com/Thatguywiththeshirt/Workshop-RedAI4All

# Rapid Fire Warm-up

## (sourced from the MIT AI Incident Database)

| Date | Company | What Happened | Failure Mode | Affected |
|---|---|---|---|---|
| 1983-09-26 | Soviet Union | Missile early-warning system flagged a false alarm; human judgment averted catastrophe | Robustness | All life on Earth |
| 2016-03-24 | Microsoft | Tay chatbot began posting racist, sexist, and anti-Semitic content within 24 hours of release | Robustness; toxic exposure | Twitter users |
| 2016-08-10 | Amazon | Resume-screening model learned historical bias, systematically downranking women | Specification failure; unfair discrimination | Female applicants |
| 2019-06-25 | Sound Intelligence | School aggression detector misclassified laughter and coughing as threatening behavior | Capability & robustness failure | Students |
| 2019-10-09 | Hive Box | Fourth-graders bypassed facial-recognition lockers using a printed photo | Security vulnerability | Hive Box customers |
| 2020-10-24 | Unknown | AI sports camera repeatedly tracked the referee's bald head instead of the ball | Robustness failure | Livestream viewers |
| 2023-01-03 | Twitter | AutoMod flagged rocket-launch images as pornographic, suspending legitimate accounts | Misclassification / false positive | Affected users |
| 2026-04-02 | KBS | During the Artemis II launch, NLP generated real time subtitles, using Korean profanity | Specification, & assurance | Editors & users |

WORKSHOP EXERCISE

# Your Turn

Three companies. Three AI deployments. One question each time: **where does this break, and who gets hurt?**

## Case 1

**Setup:** A company wants to automate procurement

**Twist:** The AI system was developed over multiple years and implemented in late 2019.

**Reveal:** $881 million loss in 2021. The model kept buying homes the market had already abandoned.

## Case 2

**Setup:** An organization wants to increase capture rates.

**Twist:** They don't have enough quality datasets to ensure accuracy.

**Reveal:** Robert Williams, Randall Reid, Porch Woodruff, Michael Oliver — all wrongfully detained. All Black.

## Case 3

**Setup:** A company wants to streamline delivery, and deploys autonomous vehicles.

**Twist:** A software glitch causes the system to misclassify what was in front of the vehicle.

**Reveal:** Elaine Herzberg was killed in Tempe, Arizona — the first pedestrian fatality caused by an autonomous vehicle.

# Let's Keep This Going

## LinkedIn

www.linkedin.com/in/petersonchase

## GitHub

https://github.com/Thatguywiththeshirt/